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Numerisk Analys, Matematik, KTH

Lab 2, Initial Value Problem, CL1-GB

Simulation of the kinetics of a biochemical reaction

A well-known model of a biochemical enzyme-substrate reaction is

E(nzyme) + S(ubstrate) = C(omplex) = E(nzyme) + P(product)

This chemical system consists of two reversible reactions, the first having rate constants k_1 (forward) and k_2 (backward) the second with rate constants k_3 and k_4. Denote by $c_i, i = 1,2,3,4$ the concentrations of E, S, C, P. Using the mass action law for the rate-expressions the following system of ODEs can be set up

$$\begin{aligned}
\frac{dc_1}{dt} &= -k_1c_1c_2 + k_2c_3 + k_3c_3 - k_4c_1c_4 \quad c_1(0) = c_{10} \\
\frac{dc_2}{dt} &= -k_1c_1c_2 + k_2c_3 \qquad\qquad\qquad\quad c_2(0) = c_{20} \\
\frac{dc_3}{dt} &= k_1c_1c_2 - k_2c_3 - k_3c_3 + k_4c_1c_4 \qquad c_3(0) = 0 \qquad \text{(1a)} \\
\frac{dc_4}{dt} &= k_3c_3 - k_4c_1c_4 \qquad\qquad\qquad\quad c_4(0) = 0
\end{aligned}$$

The ODE-system can also be written in matrix-vector form

$$\frac{dc}{dt} = Sr(c) \quad c(0) = c_0 \quad 0 \le t \le t_{end} \quad \text{(1b)}$$

where c is the column vector of concentrations, S the stoichiometric matrix, $r(c)$ the column vector of rate expressions

$$r(c) = (k_1c_1c_2,\ k_2c_3,\ k_3c_3,\ k_4c_1c_4)^T \qquad (2)$$

and c_0 the column vector of initial values.

1) What is the dimension of the matrix S ? Find and write out the matrix S.

As seen from the ODEs the system is nonlinear and, depending on the rate constants, stiff or nonstiff.

2) Explain why the system is nonlinear and explain what is meant by a stiff ODE-problem.

Your answers to 1) and 2) can either be given as comments in your program or as a parts of your report.

3) Because of mass conservation there will be linear relations between the concentrations: $c_2 + c_3 + c_4 = c_{20}$ and $c_1 + c_3 = c_{10} \quad t \geq 0$ (3)

For pass with excellence you are asked to show these relations in your report.

It is important that mass conservation is fulfilled also in the numerical solution of the ODE-system. In your program, pick out some time-points in the interval $[0, t_{end}]$, form the residuals $c_2 + c_3 + c_4 - c_{20}$ and $c_1 + c_3 - c_{10}$ and observe how well the relations (3) are fulfilled.

Your task is now to simulate the kinetics of the reaction system by solving numerically the IVP (1) and compute graphs of the four concentrations as functions of time. For your help you have the ODE-solvers ode23 and ode23s. Compare the performance of these two MATLAB-functions by noting how many timesteps are needed to reach the endpoint t_{end}.

The simulation is to be performed with the following different values of the parameters k_1, k_2, k_3, k_4 :

K1: $k_1 = 1, k_2 = 1, k_3 = 10, k_4 = 10$

K2: $k_1 = 100, k_2 = 1000, k_3 = 10, k_4 = 1.$

In both cases the initial values $c_{10} = 1, c_{20} = 0.1$ should be used and for each of K1 and K2 t_{end} should be chosen so that steady-state (equilibrium) has been reached by inspection of the graphs of the concentrations as function of time